

Question I is about the multi various chemistry of boron

1. Boron is, due to its structural diversity, to its complex chemistry and its different applications (boron compounds in detergents, glasses and boron fibres in aviation industry) a very interesting element.

The synthesis of the element can be done in different ways. Amorphous boron of minor purity may be obtained by the reaction of diboron trioxide with elemental magnesium. Very pure boron is generated by chemical vapour deposition (CVD) from a mixture of boron trichloride and hydrogen at 1300°C on tungsten wires.

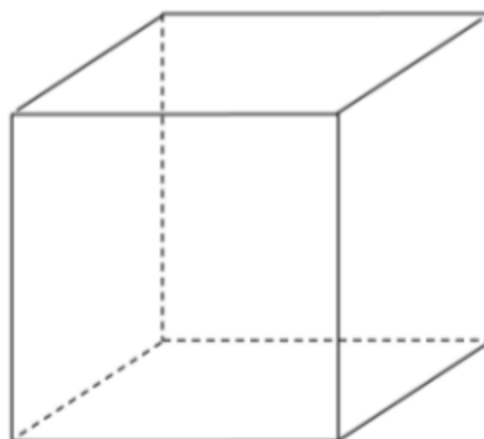
a) Write balanced chemical equations for the two-synthesis described above.

b) Which element in the neighbourhood (periodic table) has the greatest similarity to boron? Give reasons for your answer!

2. Nowadays more than 200 binary boron-metal-compounds are well known, which show a great variety in stoichiometry and in structure. Many of these metal borides are extremely hard, chemically inert and heat-resistant compounds, whose applications reach from manufacturing of thermo elements, from coating of turbine paddles, of rocket tops and of heat shields down to the use as high temperature electrodes.

Beryllium boride shows a so-called anti fluorite structure. The boron atoms thereby form a cubic closest packed lattice, in which all the tetrahedral vacancies are occupied by beryllium atoms. The shortest distance between two boron atoms has a length of 329 pm.

a) Draw the boron atoms into the elemental cell on the right side. Draw also one tetrahedron in this cell in the centre of which a beryllium atom is located.



b) How many of the tetrahedral gaps do exist? What is the empirical formula of the compound? How many formula units are there in the elemental cell?

c) What are the coordination numbers of boron and of beryllium?

d) Calculate the lattice parameter a_0 of the elemental cell!

e) Calculate the density of the boride!

f) How long is the bond between B and Be?

g) Some borides contain isolated B_2 -units. Draw the MO-scheme of B_2 !

3. Concerning the boron-hydrogen compounds, also called boranes, one may distinguish mainly between three categories of boron framework structures:

- closo: closed triangular polyhedrons
- nido (from lat. nidus = nest): open triangular polyhedrons
- arachno (from gr. arachniou = spider web): net shaped polyhedral structures

Using the so called rule of Wade one may calculate the number of framework electrons which are available for a certain structure. From this number one may determine the structure category:

number of framework electrons = sum of the valence electrons of all framework atoms (boron atoms)
 + donor electrons of substituents and ligands
 + electrons from additional charges
 – 2 electrons per main group framework atom (corresponding to the terminal binding or lone electron pair)

number of framework electrons structure

$2n + 2$ closo

$2n + 4$ nido

$2n + 6$ arachno

n = number of boron atoms

Write into the following table the number of framework electrons, the name of the structure and the proper systematic name of the given boranes!

formula	number of framework electrons	structure	name
B_5H_9			
B_6H_{12}			
$B_{10}H_{14}$			
$B_8H_8^{2-}$			

In order to synthesize the most simple borane [diborane(6)] there are different possibilities. On the one hand the reduction of boron trifluoride with sodium hydride and on the other hand the reaction of sodium boron hydride $NaBH_4$ with iodine, where as byproducts elemental hydrogen and sodium iodide will be formed.

- Write balanced reaction equations for the two synthesis!
- Which bond characteristic do you find in boranes generally?

Question II is An interesting cobalt complex

In the center of our considerations is the complex ion $[\text{CoCl}(\text{NH}_3)_5]^{2+}$.

- What is the oxidation number of Co in this ion?
- What is the systematic name of this ion.
- Draw the structure of the complex ion.
- Write down the structures of a complex which will derive from the above one if two additional ammonia molecules are substituted by Cl^- .
- How many isomers of the latter do exist?
- Write down the name for the isomeric relationship between them.

The complex ion $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ causes a red colour of the solution ($c = 0.0200 \text{ mol/L}$) with a transmission of $T = 7.50\%$, if one measures at 520 nm with a 1 cm-cuvette.

- Calculate the molar absorption coefficient of the complex.

$[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ may be reduced by Fe^{2+} in acidic solution giving Co^{2+} .

- Write a balanced equation for this reaction.

The reaction obeys a second order rate law:

$$-\frac{d\langle[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}\rangle}{dt} = k_{\text{exp}} \cdot \langle[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}\rangle \cdot \langle[\text{Fe}(\text{H}_2\text{O})_6]^{2+}\rangle$$

To simplify the investigation of the kinetics one may choose a many-fold concentration of $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ compared to the one of $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$.

- In this case the reaction order changes. How do you call it?
- Write down the new rate law.

In a kinetic experiment 10.0 mL of a solution of $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ are mixed with 10.0 mL of a solution of $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ ($c = 0.300 \text{ mol/L}$, large excess). The following data are recorded:

t (min)	10	20	40	60	∞
E	1.15	0.875	0.558	0.410	0.283

- Prove by a calculation that the reaction order you assumed in part i) is correct and calculate the rate constants for the rate law in j) and for the second order law. The concentration of the complex ions is proportional to $E - E_{\infty}$.

- Why does the reaction never reach absorption 0?

Question III is Potassium

Elementary potassium was the first metal synthesized by electrolysis (1807). Humphry Davy used voltaic cells and electrodes immersed into molten KOH.

3.1 Give balanced equations for the processes taking place on the cathode (1) and the anode (2).

It is little known that only a year after Davy, French chemists have produced potassium in a chemical reduction. One of the methods involved a mixture of potassium carbonate and finely powdered carbon. This mixture was heated to very high temperatures in a gun barrel. Carbon monoxide and potassium vapors were leaving the end of the barrel, where the metal could be condensed to a solid. 3

3.2 Write down the balanced equation for the reaction giving only these two products.

The chemical production of potassium had a low yield. Depending on the circumstances, a considerable amount of reddish paste was left behind in the gun barrel. Gmelin isolated an interesting compound (**X**) from this residue in 1825 by recrystallization in water. The exact structure of the byproduct **X** was determined more than one and a half century later.

X is a hydrated salt (2 mol water : 1 mol of salt) forming golden yellow crystals. Its anion has an exceptionally symmetric (shown by vibrational spectra) and stable planar cyclic structure.

The diprotic acid (**Y**) that can be formed from the salt **X** has surprisingly (at least to the superficial observer) low pK_a values: 0.8 and 2.2.

3.3 Give the pH where 95% of all dissolved acid **Y** is in the neutral acid form.

3.4 Calculate the pH of a 0.01 M solution of the salt **X** in pure water.

The key step in the structure identification of **X** was the treatment of the anhydrous salt with methyl iodide. The salt reacted with two equivalents of methyl iodide. The somewhat volatile compound that was obtained had a molar mass of 170.12 g/mol (measured in a mass spectrometer).

A number of analogous salts were also identified in the mixture in smaller quantities. The appropriate potassium salts when treated with two equivalent of MeI gave derivatives with molar masses of: 142.11 g/mol, 114.10 g/mol, 198.13 g/mol.

3.5 Give the empirical formula of Gmelin's compound **X**.

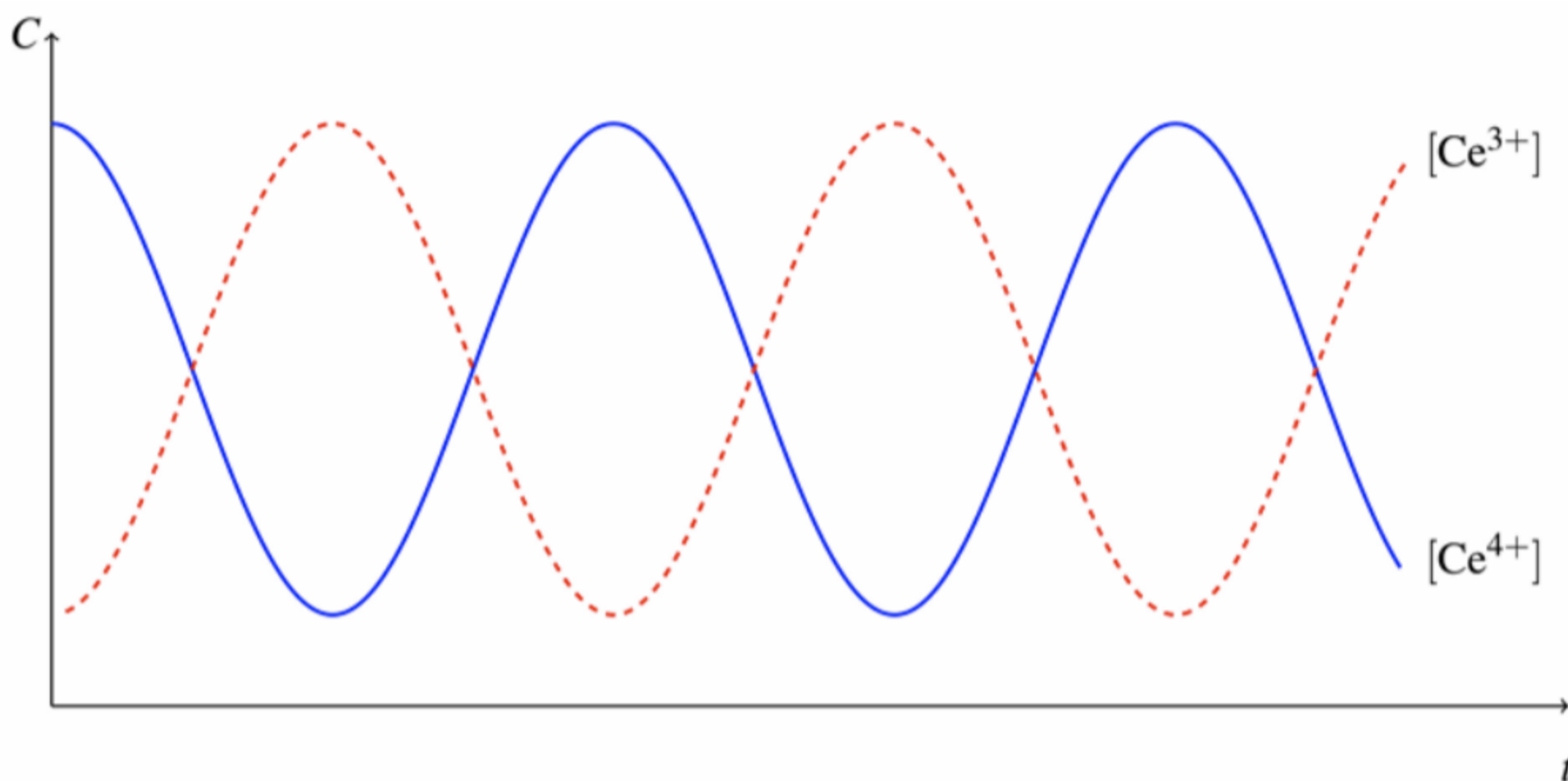
3.6 Draw the structure of the free acid **Y**.

The extraordinary stability of the series of anions in the salts analogous to **X** can be attributed to electron delocalization.

3.7 Draw one resonance structure with a significant contribution for the anion in each of the four salts.

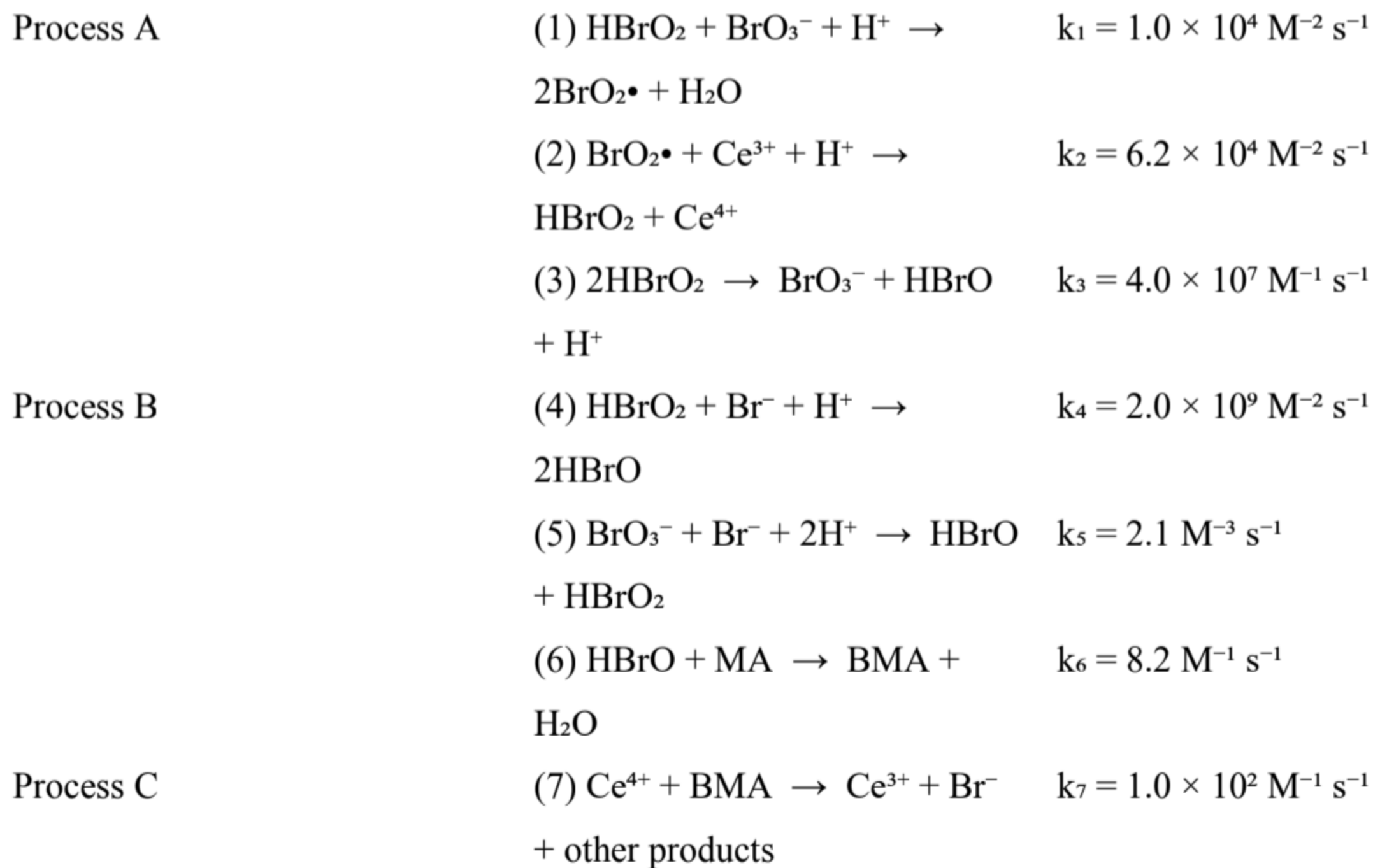
Question IV is Kinetics of the Belousov-Zhabotinsky reaction

When the chemicals KBrO_3 , $\text{CH}_2(\text{COOH})_2$ (malonic acid, – MA), $\text{Ce}(\text{SO}_4)_2$, and H_2SO_4 are mixed, the concentration, C , of Ce species and the colour of the solution oscillates between yellow (due to Ce^{4+}) and colourless (due to Ce^{3+}). A schematic picture is given below:



1. Write the overall balanced reaction equation of the Belousov-Zhabotinsky reaction (BZ reaction). Note: in this reaction, MA is oxidised to CO_2 , BrO_3^- is reduced to Br^- , and $\text{Ce}(\text{IV})$ is a catalyst.

The mechanism of the BZ reaction is shown below (BMA – $\text{CHBr}(\text{COOH})_2$):



In the BZ reaction, Process C occurs continuously, while Processes A and B alternate with each other.

Assume that:

- when Process A occurs the solution is yellow and Process B practically does not occur;
- when Process B occurs the solution is colourless and Process A practically does not occur;
- $[\text{BrO}_3^-]_0 = 0.06 \text{ M}$, $[\text{MA}]_0 = 0.1 \text{ M}$, $[\text{H}^+]_0 = 0.8 \text{ M}$, $[\text{Ce}^{4+}]_0 = 0.001 \text{ M}$;
- the concentrations of the reactants (except Ce^{4+}) and pH are maintained constant throughout the BZ reaction.

2. Using the steady-state approximation, calculate the stationary molar concentrations of HBrO_2 in Processes A and B, $[\text{HBrO}_2]_A$ and $[\text{HBrO}_2]_B$.